

Table 1:

Description	Symbol	Unit	Rat 1	Rat 2	Rat 3
<i>MTC properties</i>					
Hill constant	a^{rel}	-	0.569	0.712	0.649
Hill constant	b^{rel}	-	5.97	7.15	6.80
maximal isometric CE force	F_{CE}^{max}	N	15.5	14.1	17.0
PEE stiffness scaling factor	k_{PEE}	N/mm ²	28.7	24.1	34.8
SEE stiffness scaling factor	k_{SEE}	N/mm ²	952	1255	1050
CE optimum length	L_{CE}^{opt}	mm	13.7	14.7	13.3
PEE slack length	L_{PEE}^0	mm	15.1	15.6	14.5
SEE slack length	L_{SEE}^0	mm	28.8	29.3	25.2
Calcium dynamics activation time constant	τ_{act}	ms	55.2	57.7	41.6
Calcium dynamics deactivation time constant	τ_{deact}	ms	27.1	25.3	22.4
<i>Derived metrics</i>					
MTC length yielding maximal isometric SEE force	L_{MTC}^{opt}	mm	46.6	47.3	42.5
maximal instantaneous CE power	P_{CE}^{max}	mW	316	319	352
‘half-rise time’	t_{HRT}	ms	13.2	13.8	9.97
maximal CE shortening velocity	v_{CE}^{max}	mm/s	144	147	140
CE velocity at P_{CE}^{max}	v_{CE}^{opt}	mm/s	54.1	57.8	53.8

Table S1:

		Condition										
	Unit	1	2	3	4	5	6	7	8	9	10	11
<i>SSC parameters</i>												
Cycle frequency	Hz	1.0	2.0	3.0	4.0	5.0	3.0	3.0	3.0	3.0	5.0	4.0
FTS	-	0.50	0.50	0.50	0.50	0.50	0.80	0.65	0.35	0.20	0.80	0.6
<i>MTC shortening and lengthening times</i>												
MTC shortening time	ms	500	250	167	125	100	267	217	117	66.7	160	160
MTC lengthening time	ms	500	250	167	125	100	66.7	117	217	267	40.0	87
<i>Stimulation durations</i>												
Trial 1	ms	405	175	103	65	35	183	133	53	23	85	93
Trial 2	ms	445 ¹	205	123	85	55	213	163	73	33 ²	115	115
<i>Measured AMPO of Rat 1</i>												
Trial 1	mW	31	51	59	58	45	74	67	38	20	82	74
Trial 2	mW	34	56	69	72	66	84	77	49	25	104	87
<i>Measured AMPO of Rat 2</i>												
Trial 1	mW	29	48	57	58	45	71	62	37	18	79	70
Trial 2	mW	30	52	65	68	64	78	72	46	24	93	79
<i>Measured AMPO of Rat 3</i>												
Trial 1	mW	34	56	65	66	48	83	72	44	22	91	83
Trial 2	mW	37	62	74	82	74	91	85	56	20	116	90

¹ For conditon 1: stimulation duration of rat 3 was 455 ms. ² For conditon 9: stimulation duration of rat 3 was 23 ms. ³ For conditon 13: stimulation duration of rat 1 was 165 ms.

Table S2:

		Condition											
	Unit	1	2	3	4	5	6	7	8	9	10	11	12
<i>SSC parameters</i>													
Cycle frequency	Hz	1.0	1.5	2.0	2.5	3.0	2.0	2.0	2.0	2.0	3.0	2.5	1.5
FTS	-	0.50	0.50	0.50	0.50	0.50	0.80	0.65	0.35	0.20	0.80	0.65	0.35
<i>MTC shortening and lengthening times</i>													
MTC shortening time	ms	500	333	250	200	167	400	325	175	100	267	260	233
MTC lengthening time	ms	500	333	250	200	167	100.0	175	325	400	66.7	140	433
<i>Stimulation durations</i>													
Trial 1	ms	405	253	175	135	103	295	245	115	55	173	175	163
Trial 2	ms	455	293	205	165	123	345	275	135	75 ¹	213 ²	225	193
<i>Measured AMPO of Rat 1</i>													
Trial 1	mW	53	67	75	80	79	94	89	55	26	113	95	54
Trial 2	mW	54	47	80	86	85	99	97	63	28	121	104	62
<i>Measured AMPO of Rat 2</i>													
Trial 1	mW	49	63	73	75	76	88	83	52	24	104	88	50
Trial 2	mW	50	66	76	80	81	92	86	55	27	113	95	54
<i>Measured AMPO of Rat 3</i>													
Trial 1	mW	57	73	81	87	88	103	95	60	28	123	101	60
Trial 2	mW	47	-	89	-	88	110	-	-	32	129	-	-

¹ For conditon 1: stimulation duration of rat 3 was 455 ms. ² For conditon 9: stimulation duration of rat 3 was 23 ms. ³ For conditon 13: stimulation duration of rat 1 was 165 ms.